# Pengyu Zhang

p.zhang@uva.nl pengyu-zhang.github.io LinkedIn Google Scholar
+31 6 16144184 Amsterdam, The Netherlands

## Summary

Computer Science Ph.D. candidate (UvA) seeking Data Scientist and ML Engineer roles. I build deep learning models that fuse graph structure, text, images, and time for entity linking and recommendation. My methods beat prior state-of-the-art by up to 20% on entity linking and 6% on recommendation.

## Skills

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|---|---|
| **Programming** | Python, PostgreSQL, Bash, Git, Linux. |
| **ML & DL** | PyTorch, Hugging Face, scikit-learn, NumPy, Pandas; large language models, vision-language models, graph neural networks, contrastive learning, recommendation. |
| **Tools** | Weights & Biases, Jupyter. |
| **Languages** | Chinese (native); English (fluent, IELTS 7.0); Dutch (elementary) |

## Experience

**UvA** — **PhD Researcher, Intelligent Data Engineering Lab (INDElab)** *2022 – present*

- Full-time research on multi-modal machine learning, taking projects from idea to publication at CIKM, ECAI, ESWC, ACL, and LREC.
- Released three public benchmarks (GCL-TempEL, Graph-TempEL, Fusion Bench) and maintain the open-source code for every first-author project.

**Fusion Training** — **Hybrid-reasoning LLMs** *ACL 2026 SRW*

- Found that interleaving thinking and non-thinking training data keeps both abilities strong, and measured how the balance shifts (adding more non-thinking data steadily hurts reasoning). Released the Fusion Bench benchmark.
- Newer LLMs mix quick answers with long step-by-step reasoning to save compute, but putting both into one model makes them fight. Ran a full grid over data ratios and training orders to see what keeps both working.

**TimeRoute** — **Time-aware recommendation** *Under review*

- Raised recommendation accuracy by up to **6%** on TikTok, Amazon-Baby, and Amazon-Sports over strong baselines.
- Recommenders usually mix a user's clicks, text, and images the same way no matter when each happened. Built a model that learns which of these matter over short versus long time spans, and that cleans up noisy or missing history on its own.

**Time Imprint** — **Multi-modal entity disambiguation** *Under review*

- Raised top-match accuracy by up to **4.81%** overall, and by up to **200%** on the hardest, most look-alike cases.
- Systems often mix up near-identical records whose text and images look almost the same. Added time as an extra clue so the model can tell them apart, cutting errors most where they were worst.

**Beyond Images** — **Knowledge-graph data enrichment** *ESWC 2026*

- Raised match accuracy by up to **7%** overall, and by up to **333%** on ambiguous logos and symbols.
- Many records have missing or low-quality images, which hurts matching. Built a pipeline that finds extra images online, turns them into text with vision-language models, and writes a summary with an LLM, filling the gaps with no manual work.

**Graph-TempCZ** **Large-scale graph link prediction** *LREC 2026*

- Raised test accuracy by **5.98%** (to **92.88%**) with a GraphSAGE GNN over feature-based XGBoost baselines.
- Built the first large graph linking research papers to the software they use, with over six million mentions spanning 1959 to 2022, and checked how well the model predicts usage in later years.

**CYCLE & TIGER** **Temporally robust entity linking** *CIKM 2024 & ECAI 2024*

- CYCLE beat the best prior method by **13.9%** to **17.8%**; TIGER beat the strongest baseline by **16%** to **21%**, measured over one to three year gaps.
- Models that match text to a database lose accuracy as the database changes from year to year. CYCLE learns from those yearly changes, and TIGER also uses how records connect to each other. Both come with public benchmarks (GCL-TempEL and Graph-TempEL).

## Education

**2022 – present** **Ph.D. in Computer Science**, University of Amsterdam (UvA), The Netherlands. Supervisors: Prof. Dr. Paul Groth, Dr. Klim Zaporojets.

**2019 – 2022** **M.Eng. in Control Engineering**, Beijing University of Technology (BJUT), China. Supervisor: Prof. Dr. Yong Zhang.

## Publications

- TimeRoute: Time-Scale Experts for Multi-Modal Diffusion Recommendation. **Pengyu Zhang**, Yangqin Jiang, Klim Zaporojets, Congfeng Cao, Paul Groth. Under review.
  arXiv
- Time Imprint: Learning Time-Aware Representations in Multi-Modal Knowledge Graphs. **Pengyu Zhang**, Klim Zaporojets, Congfeng Cao, Jia-Hong Huang, Paul Groth. Under review.
  arXiv
- Fusion Training for Mathematical Generalization in Large Language Models. Congfeng Cao, **Pengyu Zhang**, Jelke Bloem. ACL 2026 (Student Research Workshop).
  Paper DOI Code
- Are a Thousand Words Better Than a Single Picture? Beyond Images: A Framework for Multi-Modal Knowledge Graph Dataset Enrichment. **Pengyu Zhang**, Klim Zaporojets, Jie Liu, Jia-Hong Huang, Paul Groth. ESWC 2026.
  Paper DOI Code YouTube Bilibili
- Graph-TempCZ: A Graph Representation of Software Mentions for Predicting Software Usage in Scientific Publications. Congfeng Cao, **Pengyu Zhang**, Jelke Bloem. LREC 2026.
  Paper DOI Code
- A Survey of Large Language Models for Data Challenges in Graphs. Mengran Li, **Pengyu Zhang**, Wenbin Xing, Yijia Zheng, Klim Zaporojets, Junzhou Chen, Ronghui Zhang, Yong Zhang, Siyuan Gong, Jia Hu, Xiaolei Ma, Zhiyuan Liu, Paul Groth, Marcel Worring. Expert Systems with Applications (ESWA) 2025.
  Paper DOI Code
- CYCLE: Cross-Year Contrastive Learning in Entity-Linking. **Pengyu Zhang**, Congfeng Cao, Klim Zaporojets, Paul Groth. CIKM 2024.
  Paper DOI Code
- TIGER: Temporally Improved Graph Entity Linker. **Pengyu Zhang**, Congfeng Cao, Paul Groth. ECAI 2024.
  Paper DOI Code